

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 30%

Annexure A

CONCEPT MAPPING

Code of Conduct:

*I **Devi Shree Babu** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

Annexure A

CONCEPT MAPPING

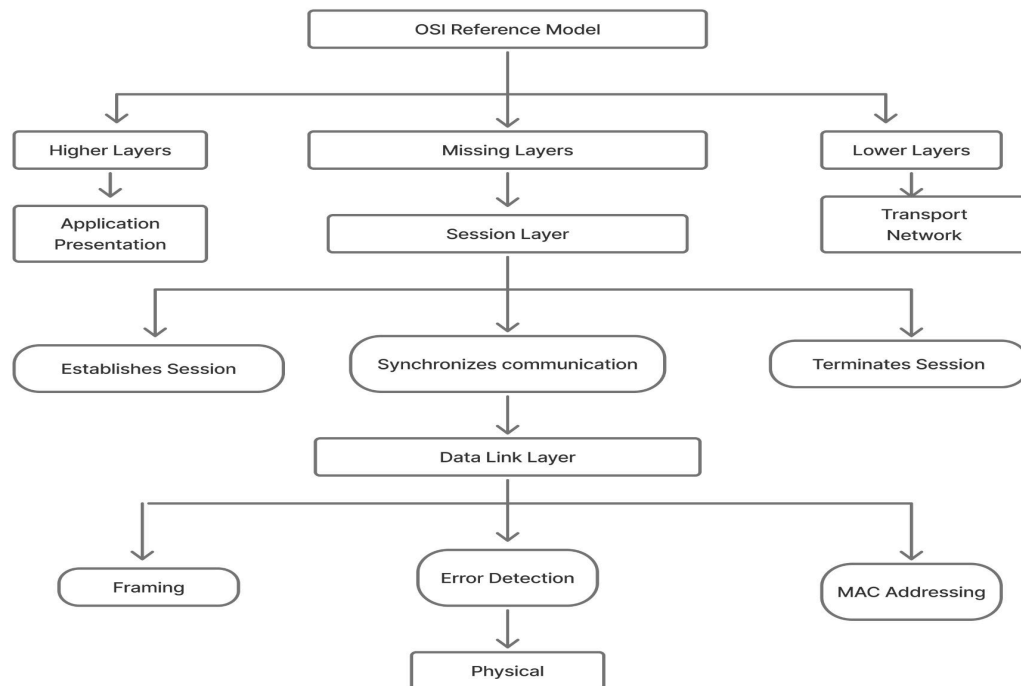
1. A partially completed concept map is given below:

Application → Presentation → _____ → Transport → Network → _____ → Physical.

Analyze the missing layers and explain how their functions are essential for reliable communication.

Answer:

Concept Map Illustrating the OSI Layers and the Functions of the Missing Layers



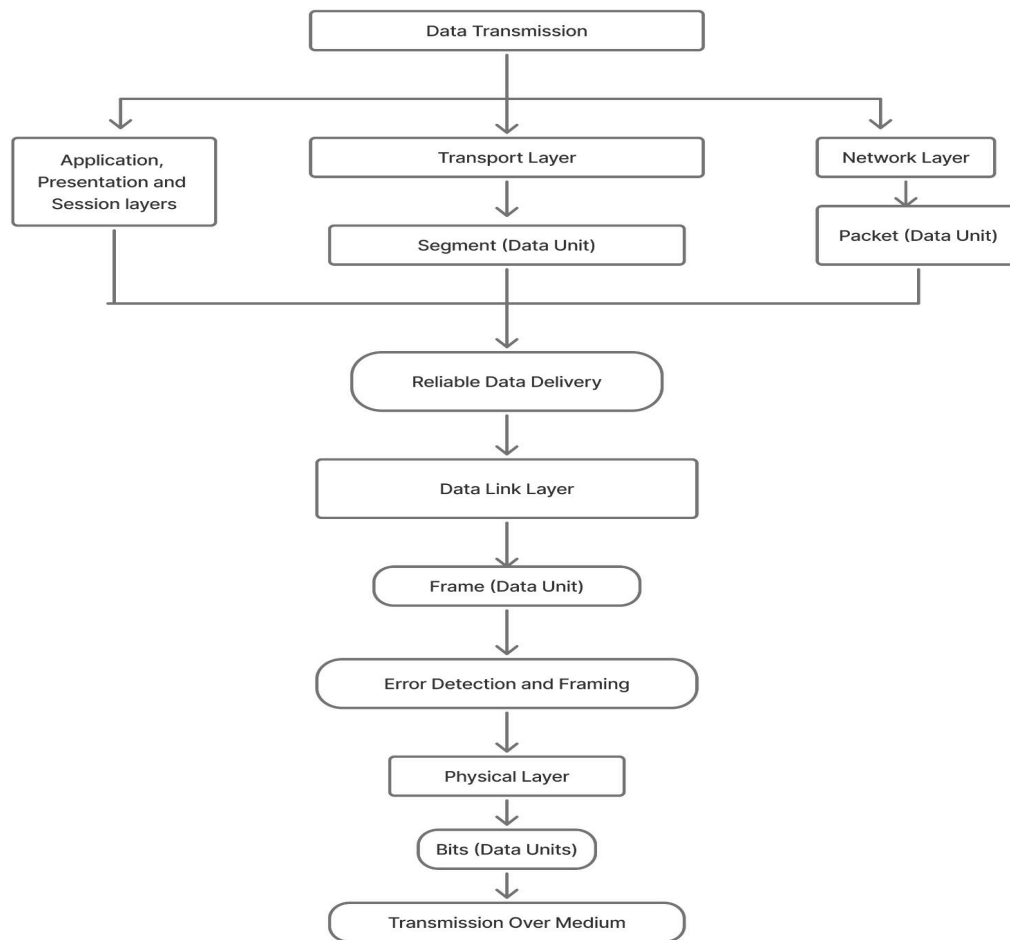
The missing layers in the given concept map are Session Layer (Layer 5) and Data Link Layer (Layer 2).

- Session Layer: Establishes, manages, synchronizes, and terminates communication sessions between applications. It ensures that communication remains organized and can recover if interrupted.
- Data Link Layer: Converts packets into frames, performs error detection, controls access to the transmission medium, and uses MAC addresses for communication between directly connected devices.

2. Construct a concept map linking data units (Segment, Packet, Frame, Bits) with their respective OSI layers and analyze how transformation of data units supports transmission.

Answer:

Concept Map of Data Unit Transformation in the OSI Model



The Transport Layer divides data into Segments for reliable end-to-end communication. The Network Layer encapsulates segments into Packets by adding logical addressing for routing. The Data Link Layer converts packets into Frames, providing framing, MAC addressing, and error detection. Finally, the Physical Layer converts frames into Bits, which are transmitted over the communication medium.

Analysis:

The transformation of data units from Segment → Packet → Frame → Bits ensures reliable communication by adding the necessary control information at each layer, enabling accurate routing, error detection, and successful transmission.

3. Given the concept map:

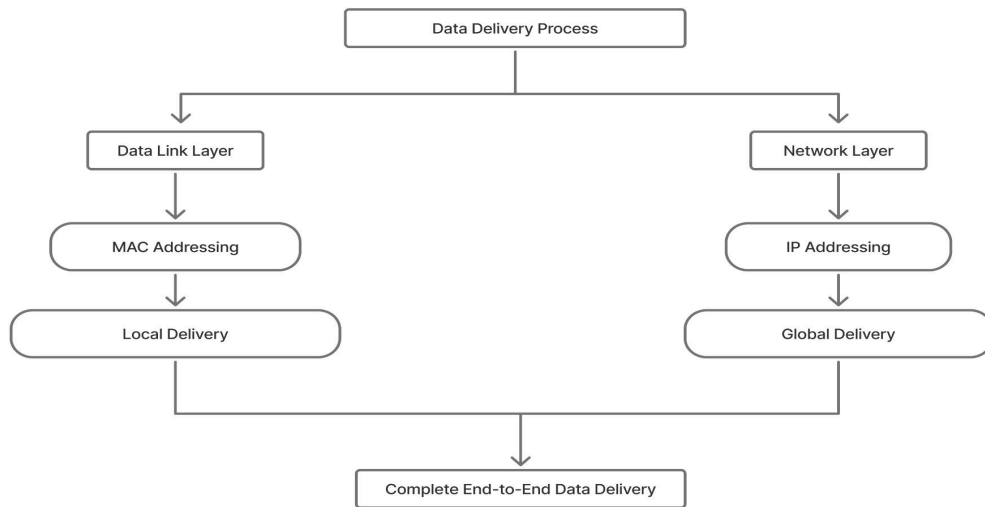
DataLink → MACAddress → LocalDelivery

Network → IP Address → Global Delivery

Explain how these relationships ensure complete data delivery across networks.

Answer:

Concept Map: MAC Addressing and IP Addressing for Complete Data Delivery



The Data Link Layer uses MAC addresses to deliver frames between devices on the same local network. The Network Layer uses IP addresses to route packets between different networks across the Internet. Together, MAC addressing provides local delivery, while IP addressing enables global delivery, ensuring that data reaches the correct destination from sender to receiver.

4. A concept map shows:

ErrorControl

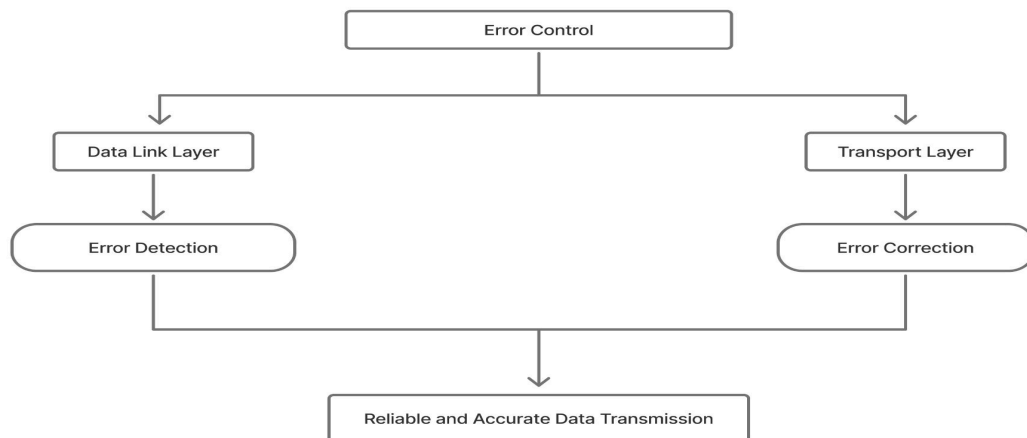
→DataLinkLayer→ErrorDetection

→ Transport Layer → Error Correction

Analyze how the interaction between these layers improves reliability in data transmission.

Answer:

Concept Map: Error Control Mechanism in the OSI Model



The Data Link Layer detects transmission errors between directly connected devices using error detection techniques. The Transport Layer ensures end-to-end reliability by performing error recovery through retransmission and acknowledgements. Together, these layers improve communication by detecting, correcting, and recovering from transmission errors, ensuring accurate and reliable data delivery.

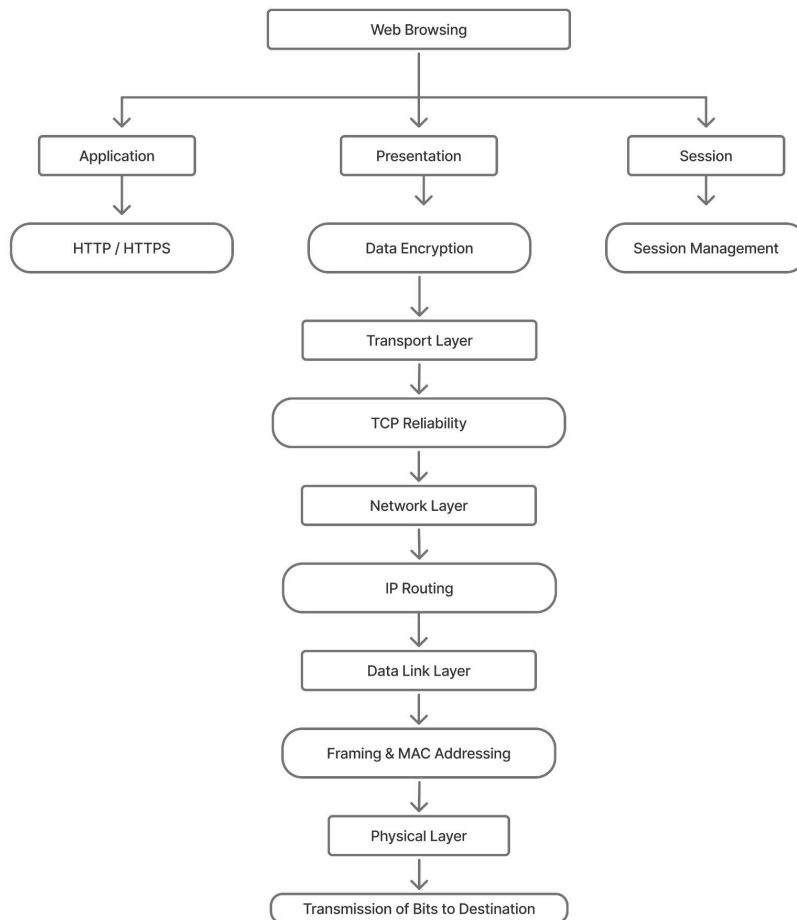
5. Develop a concept map for web browsing using OSI layers and analyze how failure in one layer affects the entire communication process.

Answer:

Concept Map: Web Browsing Process Using the OSI Model

During web browsing, each OSI layer performs a specific function, from providing web services to transmitting data. The layers work together to ensure successful communication. If any layer fails, the dependent layers cannot function properly, causing delays, data loss, or failure to load the web page. Thus, all OSI layers are essential for reliable web communication.

This is concise, directly answers the question, and fits well below the concept map without taking much space.



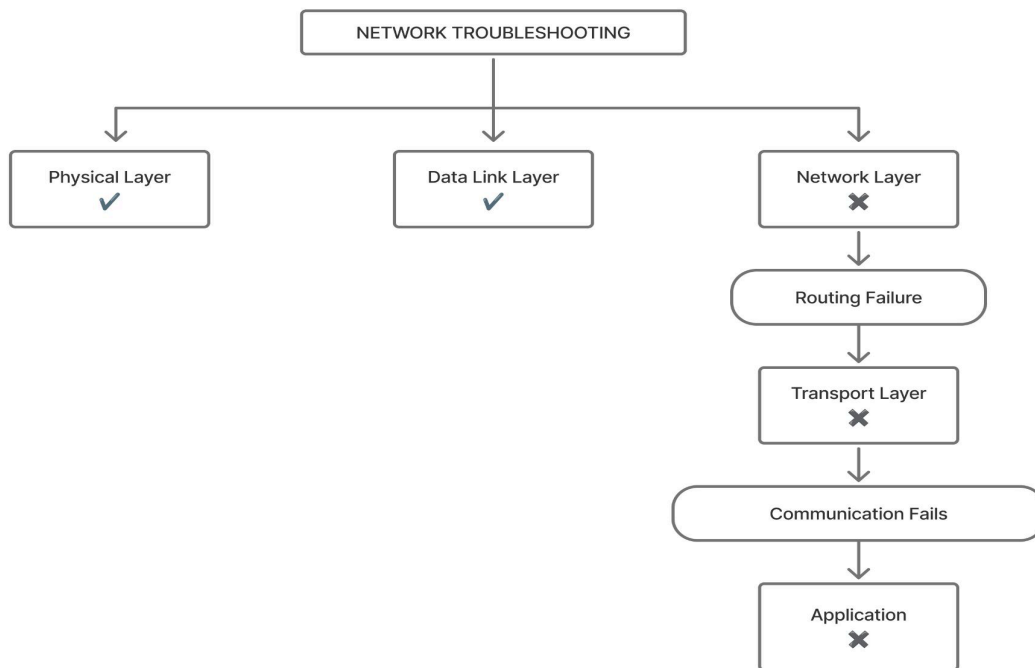
6. A network issue is represented in the concept map:

Physical ✓ → Data Link ✓ → Network ✗ → Transport ✗ → Application ✗ Analyze the problem and explain how the concept map helps in troubleshooting.

Answer:

Concept Map: OSI-Based Network Troubleshooting

The Physical and Data Link layers are functioning correctly, but the Network Layer has failed. As a result, data cannot be routed to the destination, causing the Transport and Application layers to fail. The concept map helps identify the faulty layer quickly, making network troubleshooting faster and more efficient.



RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	20%	Accurately identifies all relevant OSI layers, concepts, and relationships	Identifies most concepts with minor omissions	Identifies limited concepts; some inaccuracies	Unable to identify key concepts correctly
Concept Mapping Structure	25%	Constructs a clear, logical, and well organized concept map with proper hierarchy and connections	Concept map is mostly clear with minor structural issues	Concept map lacks clarity, organization, or proper flow	No clear structure or incorrect mapping

Linking & Relationships	20%	Clearly establishes correct relationships between layers, functions, and processes	Relationships are mostly correct with minor errors	Limited or partially correct relationships	Incorrect or missing relationships
Application & Analysis	20%	Effectively applies concepts to explain processes (e.g., encapsulation, error control) with strong reasoning	The application is correct but lacks depth in analysis	Basic application with weak explanation	Unable to apply concepts correctly
Explanation & Clarity	15%	Provides a clear, concise, and well structured explanation supporting the concept map	Explanation is understandable with minor gaps	Explanation is unclear or incomplete	No clear or meaningful explanation

